WORKED SOLUTION

STEP II 1991, Question 12

Mechanics • Hooke's Law

STEP II 1991, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q12

This question has a full guided treatment in *Beyond STEP*, Section 11.1, *Resonance*: https://beyondstep.keithmcnulty.org/ch11/resonance.html

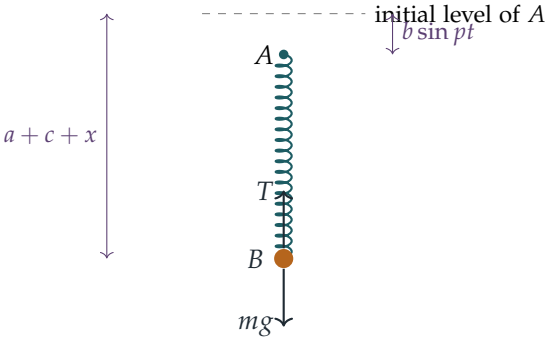
The key idea

The tension depends only on the *extension*, which is the particle's position minus the position of the moving end, minus the natural length. Write that down in terms of x and $b\sin pt$, use the initial equilibrium to replace the modulus by something involving c, and the equation of motion falls out. What follows is a forced simple harmonic oscillator. Solve it by adding a particular solution to the complementary function, and remember that "taut" means "extension never negative".

Solution

The equation of motion

Model the particle as a point of mass m on a light elastic string of modulus λ obeying Hooke's law. Measure all positions downwards from the initial position of A.



In the initial equilibrium the extension is c and the tension balances the weight:

$$\frac{\lambda c}{a} = mg, \qquad \text{so} \qquad \frac{\lambda}{ma} = \frac{g}{c} = n^2.$$

At time t the particle is at depth $a + c + x$ and the end A is at depth $b \sin pt$. While the string is taut its length is $a + c + x - b \sin pt$, so its extension is

$$e = c + x - b \sin pt,$$

and the tension is $T = \lambda e / a$. Resolving downwards for the particle,

$$\begin{aligned} m\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} &= mg - \frac{\lambda}{a}(c + x - b \sin pt) \\ &= mg - \frac{\lambda c}{a} - \frac{\lambda}{a}(x - b \sin pt) \\ &= -\frac{\lambda}{a}(x - b \sin pt), \end{aligned}$$

using $\lambda c / a = mg$. Dividing by m and using $\lambda/(ma) = n^2$,

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} = -n^2(x - b \sin pt),$$

as required. This needs $e \geq 0$: if the string went slack, the tension would be zero rather than $\lambda e / a$.

Solving for $x(t)$

Rewrite the equation as $\ddot{x} + n^2 x = n^2 b \sin pt$. The complementary function is $C \cos nt + D \sin nt$. For a particular integral try $x = K \sin pt$:

$$-p^2 K + n^2 K = n^2 b \quad \Longrightarrow \quad K = \frac{n^2 b}{n^2 - p^2},$$

which is valid because $p < n$ makes $n^2 - p^2 \neq 0$. So

$$x = \frac{n^2 b}{n^2 - p^2} \sin pt + C \cos nt + D \sin nt.$$

At $t = 0$ the particle is at its equilibrium position and at rest, so $x(0) = 0$ and $\dot{x}(0) = 0$. The first gives $C = 0$. The second gives

$$\frac{n^2 bp}{n^2 - p^2} + nD = 0 \quad \Longrightarrow \quad D = -\frac{nbp}{n^2 - p^2}.$$

Hence

$$x(t) = \frac{bn}{n^2 - p^2}(n \sin pt - p \sin nt),$$

as required.

When the string stays taut

Ask yourself What exactly has to stay positive: x, the tension, or something else?

It is the extension $e = c + x - b\sin pt$. The formula for $x(t)$ describes the motion only while $e \geq 0$, so the string stays taut throughout exactly when this holds for every $t \geq 0$. Substituting for x,

$$\begin{aligned} x - b\sin pt &= b\sin pt\left(\frac{n^2}{n^2-p^2} - 1\right) - \frac{bnp}{n^2-p^2}\sin nt \\ &= \frac{bp}{n^2-p^2}\left(p\sin pt - n\sin nt\right). \end{aligned}$$

So the necessary and sufficient condition is

$$c + \frac{bp}{n^2-p^2}\left(p\sin pt - n\sin nt\right) \geq 0 \quad \text{for all } t \geq 0,$$

or equivalently, since $n^2 - p^2 > 0$,

$$bp\left(n\sin nt - p\sin pt\right) \leq c\left(n^2-p^2\right) \quad \text{for all } t \geq 0.$$

(If e only touches 0 at isolated instants the tension is momentarily zero, but the equation of motion still holds and the string never actually goes slack. That is why the inequality is not strict.)

Now suppose $pb < (n-p)c$. For every t, since $\sin nt \leq 1$ and $-\sin pt \leq 1$, and $n, p > 0$,

$$n\sin nt - p\sin pt \leq n + p.$$

Taking $b > 0$ (the amplitude), we have $bp > 0$, so

$$bp\left(n\sin nt - p\sin pt\right) \leq bp\left(n+p\right) < (n-p)c\left(n+p\right) = c\left(n^2-p^2\right).$$

The condition holds, in fact with strict inequality, so the string remains taut throughout. (If b were allowed to be negative, the same argument works with $|b|$ in place of b, since $|n\sin nt - p\sin pt| \leq n + p$.)

Checks

- With $t = 0$: $x(0) = 0$ and $\dot{x}(0) = \frac{bn}{n^2-p^2}(np - pn) = 0$, as they should be.
- As $p \to 0$ the end A barely moves, and $x(t) \approx \frac{bn}{n^2}\left(npt - pnt\right) \to 0$. Nothing happens, as expected.
- The minimum extension can't go below $c - bp(n+p)/(n^2-p^2) = c - bp/(n-p)$, which is positive exactly when $pb < (n-p)c$.
- Numerically, a step-by-step (RK4) integration of the full Hooke's-law equation, with random m, λ, a, c, b, p, matched the formula for $x(t)$ to within 4×10^{-10}. With $n = 1$, $p = 1/\sqrt{2}$, $c = 1$ and b just 5% above $(n-p)c/p$, the extension does become negative (about -0.05) by $t = 20\,000$.

Common slips

- Using x itself as the extension. The top end moves, so the extension is $c + x - b\sin pt$.
- Forgetting that $\dot{x}(0) = 0$ forces a $\sin nt$ term in the solution. Without it the particle would start with velocity $n^2bp/(n^2 - p^2)$.
- Writing the taut condition as $x \geq -c$. That would be right only if A were fixed.

Beyond STEP

This is a forced, undamped oscillator. The response is a mix of the driving frequency p and the natural frequency n, and the factor $1/(n^2 - p^2)$ blows up as $p \to n$, which is resonance. When n/p is irrational the values of (nt, pt) modulo 2π fill out densely, so $n\sin nt - p\sin pt$ gets arbitrarily close to $n + p$. The sufficient condition $pb < (n - p)c$ is then essentially also necessary (equality is still allowed). Forced oscillations, resonance and damping are treated thoroughly in Kleppner and Kolenkow, *An Introduction to Mechanics*, and the differential-equation side in Boyce and DiPrima, *Elementary Differential Equations*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.